

Enterprise AI Digital Twin - API Documentation

Overview

The **Enterprise AI Digital Twin API** is a FastAPI-powered REST and WebSocket service providing high-performance access to the 5-Agent transformation pipeline, NetworkX graph traversal engine, Qdrant vector search, authentication, and enterprise persistence stores.

- **Base URL:** `http://localhost:8000`
 - **Interactive OpenAPI Specs (Swagger UI):** `http://localhost:8000/docs`
 - **ReDoc Documentation:** `http://localhost:8000/redoc`
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1. Authentication API (`/api/v1/auth`)

Security is implemented via **JWT (JSON Web Tokens)** using HMAC-SHA256 signatures. Authenticated endpoints require the `Authorization: Bearer <token>` HTTP header.

POST `/api/v1/auth/signup`

Registers a new enterprise user account and returns a JWT access token.

Request Body:

```
{
  "email": "executive@meridian.com",
  "password": "SecurePassword2026!",
  "name": "Enterprise Executive"
}
```

Response (200 OK):

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "user": {
    "id": 1,
    "email": "executive@meridian.com",
```

```
    "name": "Enterprise Executive"
  }
}
```

POST /api/v1/auth/login

Authenticates user credentials and issues a valid JWT Bearer access token.

Request Body:

```
{
  "email": "executive@meridian.com",
  "password": "SecurePassword2026!"
}
```

Response (200 OK):

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "user": {
    "id": 1,
    "email": "executive@meridian.com",
    "name": "Enterprise Executive"
  }
}
```

GET /api/v1/auth/me

Retrieves the current authenticated user profile using the JWT Bearer token.

Request Headers:

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Response (200 OK):

```
{
  "id": 1,
  "email": "executive@meridian.com",
  "name": "Enterprise Executive"
}
```

2. Core Transformation Pipeline API

POST /api/v1/ingest

Triggers the full **LangGraph 5-Agent Pipeline** to parse strategic directives, map value chains, generate processes, analyze role skill transitions, and assess governance compliance.

Request Body:

```
{
  "strategy_text": "Become an AI-first regional retailer within 3 years – improve ma
  "time_horizon_years": 3,
  "user_email": "executive@meridian.com"
}
```

Response (200 OK):

```
{
  "status": "success",
  "organisation_id": 1,
  "strategy_id": 1,
  "summary": {
    "value_chain_stages": 5,
    "processes": 12,
    "activities": 36,
    "roles": 8,
    "skill_transitions": 14,
    "ai_opportunities": 10,
    "governance_assessments": 10
  }
}
```

3. Graph Traversal & Executive Query API

POST /query

Performs 2-hop graph traversals over the in-memory NetworkX knowledge graph (`G_live`) and injects Qdrant vector evidence to answer executive questions (`ask: [question]`).

Request Body:

```
{
  "question": "ask: what should we optimize first and why?",
  "entity_type": "process",
  "entity_id": 2
}
```

Response (200 OK):

```
{
  "question": "ask: what should we optimize first and why?",
  "answer": "### Executive Transformation Advisory\n\nBased on our Enterprise Digital  
Transformation  
Initiative, we recommend  
focusing on the Supply Chain & Logistics  
area for optimization. This area  
has the highest potential for  
improvement, with a node count of 307  
and an edge count of 250. The  
sample nodes are as follows:
```

4. Simulation & Scenario Engine API

POST /simulate

Executes hypothetical simulation diffs against the Digital Twin graph (`automate_activity` or `delay_initiative`) and evaluates node state changes.

Request Body:

```
{
  "action": "automate_activity",
  "target_id": 4,
  "params": {
    "automation_level": 0.85
  }
}
```

Response (200 OK):

```
{
  "status": "simulation_complete",
  "target_entity": "activity:4",
  "diff": {
    "changed_nodes": [
      {
        "id": "activity:4",
        "status": "automated",
        "human_labor_impact": -0.85
      }
    ],
    "dimmed_nodes": [
      {
        "id": "role:2",
        "status": "skill_declining"
      }
    ]
  },
  "narrative": "Simulation scenario 'automate_activity' on 'activity:4' executed. Gr"
}
```

5. Human-In-The-Loop (HITL) Governance API

POST /agent/hitl_response

Allows AI compliance officers to manually review, approve, or reject high-risk AI opportunity proposals flagged during pipeline execution.

Request Body:

```
{
  "interrupt_id": "14",
  "decision": "approve"
}
```

Response (200 OK):

```
{
  "status": "success",
  "assessment_id": 14,
  "signoff_status": "approved",
  "message": "Governance signoff approved successfully."
}
```

6. Workspace & Chat History API

GET /chats

Retrieves saved chat sessions and execution thinking steps. Optional filter by `user_email` .

Response (200 OK):

```
[
  {
    "id": "session-1785821583561",
    "title": "Become an AI-first regional retailer...",
    "timestamp": 1785821583561.0,
    "messages": [
      {
        "id": "u-1785821584510",
        "role": "user",
        "text": "what should we optimize first",
        "progress": 100
      }
    ]
  }
]
```

POST /chats

Saves or updates a chat session with full message history and thinking steps.

DELETE /chats/{session_id}

Deletes a saved chat session by ID.

7. NetworkX Knowledge Graph & Metadata APIs

GET /api/v1/graph

Exports the full NetworkX knowledge graph structure as JSON (`nodes` and `links`) for frontend visual canvas rendering.

GET /api/v1/organisations

Lists all enterprise organizations registered in the platform.

8. HTTP Status Codes

Status Code	Description
200 OK	Request completed successfully.
400 Bad Request	Invalid payload, duplicate email, or missing required parameters.
401 Unauthorized	Invalid JWT token signature, expired token, or invalid credentials.
404 Not Found	Requested entity (user, session, organization, or node) not found.
500 Server Error	Pipeline execution exception or database connection failure.